

SPECIFICATION FOR UNUSED MINERAL INSULATING OILS FOR TRANSFORMERS AND SWITCHGEAR

SECTION ONE — GENERAL CONSIDERATIONS AND TEST METHODS

1. Scope

- 1.1 This standard covers specifications and test methods for unused mineral insulating oils, as delivered, intended for use in transformers, switchgear and similar electrical equipment in which oil is required as an insulant or for heat transfer. These oils are obtained by distillation and refining of petroleum. Oils with and without additives are within the scope of this standard.

- 1.2 This standard does not apply to mineral insulating oils used as impregnants in cables or capacitors nor to hydrocarbon insulating liquids obtained by synthesis.

Note. — Oils complying with the requirements of this standard, of the same class and containing no additives are considered to be compatible with one another and can be mixed in any proportion. This does not necessarily apply to oils containing additives and where the user wishes to mix such oils a check should be made to ensure that the mixture meets the requirements of this standard.

2. Definitions

2.1 Additive

A suitable substance, which is deliberately added to an insulating liquid in small proportion in order to improve certain characteristics.

2.2 Anti-oxidant

An additive incorporated in an insulating liquid to reduce or delay its degradation by oxidation.

2.3 Pour point depressant

An additive that enables the pour point of a mineral insulating oil to be lowered.

2.4 Uninhibited oil

A mineral insulating oil, containing no anti-oxidant; but which may contain other additives.

Note. — In certain countries, oil containing up to 0.08% by mass of 2,6-di-tert-butyl-paracresol (DBPC) or 2,6-di-tert-butyl-phenol (DBP) are considered as uninhibited oils.

2.5 Inhibited oil

A mineral insulating oil which contains an anti-oxidant.

Note. — In certain countries, an inhibited mineral insulating oil is defined as a mineral insulating oil containing at least 0.15% by mass, but no more than 0.40% by mass of 2,6-di-tert-butyl-paracresol (DBPC) or 2,6-di-tert-butyl-phenol (DBP).

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3. Classification

For the purpose of this standard three grades of oil are considered; Class I, Class II, Class III, based on the values of certain physical properties, namely; flash point, pour point and viscosity at 40 °C and at a specified lower temperature.

The letter A following the number of the class indicates the presence of an anti-oxidant.

4. Identification and general delivery requirements

4.1 Oil is normally delivered in road or rail tank wagons or in drums, which shall be especially cleaned for this purpose.

4.2 Oil drums and containers of oil samples shipped by the oil supplier shall carry at least the following markings:

- supplier's designation;
- class of oil.

4.3 Each oil delivery shall be accompanied by a document from the supplier specifying at least: supplier's designation, oil class and the function of any additive present.

If an anti-oxidant is present its generic nature shall be indicated.

Where a pour point depressant is included in an oil formulation this shall be done by agreement between purchaser and supplier: in this case the generic nature of the depressant shall be made known to the purchaser.

Note. — The purchaser may require that no major changes in crude oil type or refining process may be made without the purchaser being informed.

5. Sampling

Sampling shall be carried out in accordance with the procedure described in IEC Publication 475.

As a general rule, oil samples should be taken where the oil is presumed to be more contaminated, for example, at the lowest point of the container; however, contaminants at the surface or in the body of the oil should not be overlooked.

6. Specification

6.1 Mineral insulating oil complying with this standard shall meet the limiting values specified in Table I or in Table II when tested in accordance with the methods specified in Clause 7.

These limiting values apply only to unused oil, as delivered, prior to its introduction into apparatus.

6.2 Precision data given in the specified methods are to be used solely as a guide to the expected agreement between duplicate measurements of the property and are not to be regarded as tolerances applicable to the limits specified in Tables I and II.

Note. — In some applications, and particularly for special climatic conditions, oils with other characteristics may be used. Such oils may not comply with some of the requirements of this standard.

7. Test methods

7.1 Appearance

Appearance shall be evaluated by examining a representative sample of the oil in transmitted light under a thickness of approximately 10 cm and at ambient temperature.

7.2 Density

Any recognized test method may be used. The method to be used in cases of dispute shall be that given in ISO Standard 3675.

Density shall be measured at 20 °C or converted to 20 °C by the formula:

$$\rho_{20} = \rho_t [1 + X (t - 20)]$$

where:

ρ_{20} = density at 20 °C

t = temperature in degree Celsius

ρ_t = density measured at temperature t

X = correction factor = $65 \cdot 10^{-5}$ (approximate value)

For greater accuracy, the correction factor listed in ISO Recommendation R/91 should be used.

7.3 Kinematic viscosity

Kinematic viscosity shall be measured by the method given in ISO Standard 3104.

7.4 Flash point

Flash point shall be determined according to ISO Standard 2719.

7.5 Pour point

Pour point shall be determined according to ISO Standard 3016.

7.6 Interfacial tension

Interfacial tension shall be determined in accordance with ISO Standard 6295.

7.7 Neutralization value

7.7.1 Scope of the test method

This method is suitable for the determination of the total acidity of unused and used mineral insulating oils.

7.7.2 Reagents

- Standard alcoholic potassium hydroxide (KOH) solution 0.1 mol/l.
- Toluene, sulphur-free.
- Azeotropic ethanol (boiling point 78.2 °C).
- Standard hydrochloric acid solution (HCl) 0.1 mol/l.
- Alkali blue indicator solution: 2 g of alkali blue 6 B are dissolved in 100 ml of azeotropic ethanol containing 1 ml of hydrochloric acid 0.1 mol/l. After 24 h, carry out a titration to check

whether the indicator has been sufficiently sensitized. The indicator is satisfactory if the colour changes distinctly from blue to a red comparable to that of a 10% solution of cobalt nitrate $[\text{Co}(\text{NO}_3)_2 \cdot 6 \text{H}_2\text{O}]$. Should sensitization be insufficient repeat the addition of 0.1 mol/l HCl and check again after 24 h. Continue until sensitization is satisfactory. Filter and store in a brown bottle in the dark.

7.7.3 Procedure

Weigh 20 g of the sample to the nearest 0.05 g into a 250 ml stoppered conical flask.

To a mixture of 60 ml of toluene and 40 ml of ethanol in a second flask add 2 ml of indicator solution. Neutralize the solution with 0.1 mol/l KOH solution until a red colour comparable to that of a solution of 10% cobalt nitrate $[\text{Co}(\text{NO}_3)_2 \cdot 6 \text{H}_2\text{O}]$ is obtained and persists for at least 15 s.

Add this solution to the sample, swirl and immediately titrate at a temperature not above 25 °C with 0.1 mol/l KOH solution to the above end point.

The neutralization value (NV) is calculated from the expression:

$$NV = \frac{V \times N \times 56.1}{P}$$

where:

V = number of millilitres of 0.1 mol/l KOH used in the titration

N = molarity of the KOH solution

P = mass of the oil sample in grams

7.7.4 Precision

Under consideration.

7.8 Corrosive sulphur

Corrosive sulphur shall be determined according to ISO Standard 5662.

7.9 Water content

Water content shall be determined according to IEC Publication 733.

7.10 Anti-oxidant additives

Methods for the detection and quantitative determination of specified anti-oxidant additives are described in IEC Publication 666.

7.11 Oxidation stability

7.11.1 The oxidation stability of uninhibited mineral insulating oils shall be determined by the method described in IEC Publication 74.

7.11.2 The procedure described in IEC Publication 474 may be used to check the continuity of the oxidation stability of an inhibited oil.

7.12 Breakdown voltage

7.12.1 Breakdown voltage shall be determined by the test described in IEC Publication 156.

7.12.2 When breakdown voltage is to be determined on a treated oil sample the following procedure shall be used to remove moisture and particulate contaminants present in the original sample:

A sufficient oil volume is heated to approximately 60 °C, then filtered hot under a vacuum of about 2.5 kPa through a sintered glass filter of grade P10 porosity (ISO Standard 4793).

The filtrate is cooled in a desiccator and used immediately to determine breakdown voltage.

Note. — Filtration can be done equally well at ambient temperature, but under lower pressure (e.g. 0.5 kPa).

7.13 Dissipation factor

7.13.1 This property shall be measured at 90 °C using the method described in IEC Publication 247.

Note. — When oil as delivered shows a dissipation factor greater than the limit specified in Table I and Table II, a repeat measurement made on a sample treated as indicated in Sub-clause 7.12.2 will show if contaminants are removed by a relatively mild treatment.

SECTION TWO — INDIVIDUAL SPECIFICATIONS

TABLE I

Specification for uninhibited mineral insulating oils

Property	Test method, sub-clause of Section One of this publication	Permissible values for measured characteristics		
		Class I	Class II	Class III
Kinematic viscosity, mm ² /s at 40 °C a) -15 °C -30 °C -40 °C	7.3	≤16.5 ≤800	≤11.0 ≤1 800	≤3.5 ≤150
Flash point, °C	7.4	≥140	≥130	≥95 b)
Pour point, °C	7.5	≤-30	≤-45	≤-60
Appearance	7.1	Clear, free from sediment and suspended matter		
Density, kg/dm ³ at 20 °C	7.2	≤0.895 c)		
Interfacial tension, N/m at 25 °C	7.6	d)		
Neutralization value, mg KOH/g	7.7	≤0.03		
Corrosive sulphur	7.8	Non-corrosive		
Water content, mg/kg	7.9	e)		
Anti-oxidant additives	7.10	Not detectable f)		
Oxidation stability: g) neutralization value, mg KOH/g sludge, % by mass	7.11.1	≤0.40 ≤0.10		
Breakdown voltage: as delivered, kV after treatment, kV	7.12	≥30 ≥50 h)		
Dissipation factor, at 90 °C and 40 Hz to 60 Hz	7.13	≤0.005		

a) Corresponding viscosity values at 20 °C are approximately 40 mm²/s for Class I oil, 25 mm²/s for Class II oil and 6 mm²/s for Class III oil.

b) In certain countries, a flash point higher than 100 °C is normally required due to legal or safety regulations.

c) A maximum limit for density is specified in the attempt of minimizing the risk of ice crystals floating in the oil where oil-insulated equipment is exposed to very low temperatures.

d) No requirement for interfacial tension is included in this specification. However, in some countries, it is usual for standards to specify a minimum value, in this case the interfacial tension should be not less than $40 \cdot 10^{-3}$ N/m.

e) No requirement for water content on delivery is included as it is usual in most applications to process the oil before use. However, in some countries, it is usual for standards to specify maximum water content on delivery, in this case the water content should be less than 30 mg/kg in bulk delivery and less than 40 mg/kg in drum delivery.

f) No antioxidant shall be included in uninhibited oils (see also the note in Sub-clause 2.4).

g) The determinations on the oxidized oil are limited to the neutralization value and to the sludge, however in certain countries it is usual to determine also the dielectric dissipation factor. In this case the maximum value of the dielectric dissipation factor will be established by agreement between purchaser and supplier.

h) This test is only required when the oil sample taken on delivery shows breakdown voltage less than 30 kV and it is desired to show that contaminants may be effectively removed by a relatively mild treatment.

TABLE II
Specification for inhibited mineral insulating oils

Property	Test method, sub-clause of Section One of this publication	Permissible limiting values for measured characteristics		
		Class IA	Class IIA	Class IIIA
Kinematic viscosity, mm ² /s at 40 °C ^{a)} -15 °C -30 °C -40 °C	7.3	≤ 16.5 ≤ 800	≤ 11.0 ≤ 1 800	≤ 3.5
Flash point, °C	7.4	≥ 140	≥ 130	≥ 150
Pour point, °C	7.5	≤ -30	≤ -45	≥ 95 ^{b)} ≤ -60
Appearance	7.1	Clear, free from sediment and suspended matter ≤ 0.895 ^{c)} ^{d)} ≤ 0.03 Non-corrosive ^{e)} ^{f)} ^{h)} ≥ 30 ≥ 50 ⁱ⁾ ≤ 0.005		
Density, kg/dm ³ at 20 °C	7.2			
Interfacial tension, N/m at 25 °C	7.6			
Neutralization value, mg KOH/g	7.7			
Corrosive sulphur	7.8			
Water content, mg/kg	7.9			
Anti-oxidant additives	7.10			
Oxidation stability ^{g)}	7.11.2			
Breakdown voltage: as delivered, kV	7.12			
after treatment, kV				
Dissipation factor, at 90 °C and 40 Hz to 60 Hz	7.13			

- a) Corresponding viscosity values at 20 °C are approximately 40 mm²/s for Class IA oil, 25 mm²/s for Class IIA oil and 6 mm²/s for Class IIIA oil.
- b) In certain countries, a flash point higher than 100 °C is normally required due to legal or safety regulations.
- c) A maximum limit for density is specified in the attempt of minimizing the risk of ice crystals floating in the oil where oil-insulated equipment is exposed to very low temperatures.
- d) No requirement for interfacial tension is included in this specification. However, in some countries, it is usual for standards to specify a minimum value, in this case the interfacial tension should be not less than $40 \cdot 10^{-3}$ N/m.
- e) No requirement for water content on delivery is included as it is usual in most applications to process the oil before use. However, in some countries, it is usual for standards to specify maximum water content on delivery, in this case the water content should be less than 30 mg/kg in bulk delivery and less than 40 mg/kg in drum delivery.
- f) Type and content of the anti-oxidant additive should be as stipulated between purchaser and supplier (see also the note in Sub-clause 2.5).
- g) In the case of inhibited oils only the induction period is determined, however, in certain countries it is usual to determine also the dielectric dissipation factor. In this case, the maximum value of the dielectric dissipation factor will be established by agreement between purchaser and supplier.
- h) No limit is specified to date. For guidance only, oils known to give satisfactory performance in transformers in service typically have induction periods in excess of 120 h.
- i) This test is only required when the oil sample taken on delivery shows breakdown voltage less than 30 kV and it is desired to show that contaminants may be effectively removed by a relatively mild treatment.